

Assignment 1 – Case Study

A regional bank has experienced a significant increase in ATM fraud incidents over the past year. These incidents include:

- **Skimming** attacks that steal card data.
- **Jackpotting** attacks that force ATMs to dispense cash.
- **Network intrusions** that manipulate transaction data.

Analyze the existing "**ATM Technical Requirements**" document:

- **Identify** vulnerabilities that these attacks exploit, and propose updated requirements to mitigate these threats.
- Briefly **discuss** the importance of ATM security in today's financial landscape.
- Introduce the concept of evolving security requirements and the need for continuous improvement.
- **Analyze** the provided document and identify potential security gaps or outdated requirements. **Focus** on areas where technology has advanced or where new threats have emerged.
- **Research** current trends and best practices in ATM security (*Explore academic papers, industry reports, and security standards (e.g., Payment Card Industry Data Security Standard – PCI DSS).*)
- **Create** a new or updated ATM technical requirement document. This document should address the identified gaps and incorporate the cutting-edge security measures from their research, justify each new requirement and explain its importance.

The new document should include sections like:

1. Prevention (Stop attacks before they happen)
 - i. Enhanced Authentication and Authorization
 - ii. Secure Software and Hardware
 - iii. Tamper-Proof Hardware and Software
 - iv. Secure Network Communication Protocols
 - v. Compliance and Standards
2. Detection (Identify threats in real-time or early stages)
 - i. Advanced Threat Detection and Prevention
 - ii. Intrusion Detection and Prevention Systems (IDPS)
 - iii. Network Security
 - iv. Real-Time Monitoring and Incident Response
3. Response & Recovery (Act after detecting or suffering an attack)
 - i. Incident Response and Forensics
 - ii. Provides structured processes for handling attacks, collecting evidence, and learning from incidents.

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- iii. Data Protection and Privacy
- iv. While partially preventive, it also ensures data remains protected even after an incident, minimizing impact.

Plagiarism & AI Use Policy:

Criteria	Allowed Limit	Penalty
Plagiarism (Text Similarity)	Up to 15%	Above 15%: Marks deducted; excessive similarity (>50) will result in 0 marks.
AI-Generated Content	Up to 50%	Above 50% AI content: 0 marks
Duplicate Submission	Not allowed	If two or more students submit identical or near-identical content, all involved will receive zero and be reported under academic misconduct
Turnitin Use	Yes	Checked for both plagiarism and AI usage

Note: Use AI for inspiration only – the core analysis and reflection must be your own.

Marking Criteria (Total: 25 Marks, Weightage: 5)

Section	Details	Marks
1. Threat Analysis and Vulnerability Identification	Identification of how skimming, jackpotting, and network intrusions work, and the vulnerabilities they exploit in existing ATM systems.	5 marks
2. Importance of ATM Security	Brief yet insightful discussion on the significance of ATM security in today's financial environment, customer trust, and regulatory compliance.	2 marks
3. Concept of Evolving Security Requirements	Explanation of why ATM security cannot be static, the evolving threat landscape, and the importance of continuous updates.	2 marks
4. Gap Analysis of Existing ATM Technical Requirements	Identification of outdated or missing elements in the existing document, with examples and rationale.	4 marks
5. Research on Best Practices & Standards	Use of credible sources like PCI DSS, NIST guidelines, industry reports, and recent case studies to support the proposed security upgrades.	4 marks
6. New/Updated ATM Technical Requirements Document	Clear, well-structured, and detailed security document with three major sections: Prevention, Detection, and Response & Recovery. Each requirement should be explained and justified.	6 marks
7. Presentation, Clarity, and Technical Writing	Professional formatting, clarity of language, logical organization, and quality of argumentation.	2 marks

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